ONLINE SURVEY PLATFORM

A Full-Stack Style JavaScript Mini-Project

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[full stack development]

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PROJECT OBJECTIVE & SCOPE

OBJECTIVE

To build a simple, responsive, and dynamic survey platform directly within the browser using Core Web Technologies.

SCOPE

- Collect user data through a multi-input interactive form.
- Dynamically calculate and display results in real-time.
- Demonstrate fundamental JavaScript concepts without relying on external libraries (like React or Vue).

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TECH STACK

HTML5

For semantic structure and built-in form validations (e.g., required, min, max, type="email").

CSS3

Pure CSS for responsive design, Flexbox layouts, interactive pseudo-classes (:hover, :focus), and clean UI boundaries.

VANILLA JAVASCRIPT

Core logic, DOM manipulation, array mappings, and dynamic UI updates without third-party dependencies.

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PLATFORM FEATURES & FUNCTIONALITIES

COMPREHENSIVE DATA COLLECTION

Utilizes multiple input variables (Text, Email, Radio, Checkboxes, Dropdowns, Sliders, Textareas).

REAL-TIME FORM VALIDATION

Enforces constraints before submission.

DYNAMIC RESULTS DASHBOARD

Automatically converts submitted data into structured tables.

STATISTICAL COMPUTATION

Calculates aggregated metrics (Total Responses, Top Language, Average Ratings, Recommendation Percentages).

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ARRAY OF OBJECTS DATA MODEL

MECHANISM

User payloads are captured, normalized, and stored as JavaScript Objects.

STORAGE

These objects are continuously pushed into a primary global array (surveyResponses).

ADVANTAGE

Mimics a backend database structure natively in the browser, making data mapping and retrieval highly efficient.

```javascript
// Initialize the global array
const surveyResponses = [];

// Example survey response object
const newResponse = {
  id: 1,
  email: "user@example.com",
  feedback: "Great platform!"
};

// Push the new response into the array
surveyResponses.push(newResponse);

// Another example
const anotherResponse = {
  id: 2,
  email: "another@user.com",
  feedback: "Easy to use."
};
surveyResponses.push(anotherResponse);

// console.log(surveyResponses);
// Output:
// [
//   { id: 1, email: "user@example.com", feedback: "Great platform!" },
//   { id: 2, email: "another@user.com", feedback: "Easy to use." }
// ]
```

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JAVASCRIPT ITERATORS & CONSTRAINTS

FOREACH LOOPS

Extensively used to map array sets into dynamic HTML table rows.

```
const data = [{id: 1, name: 'Alice'}, {id: 2,
name: 'Bob'}];
const tableBody =
document.querySelector('tbody');

data.forEach(item => {
  const row = document.createElement('tr');
  row.innerHTML = `<td>${item.id}</td>
<td>${item.name}</td>`;
  tableBody.appendChild(row);
});
```

FOR LOOPS

Used as computation engines to aggregate metric data across all stored responses.

```
const responses = [
  {score: 5}, {score: 4}, {score: 5}
];
let totalScore = 0;

for (let i = 0; i < responses.length; i++) {
  totalScore += responses[i].score;
}

const averageScore = totalScore /
responses.length;
// console.log(`Average Score:
${averageScore}`);
```

FOR...IN LOOPS

Used to traverse object properties (like language counts and interest areas) to find the most popular choices.

```
const languageCounts = {
  JavaScript: 15,
  Python: 10,
  Java: 8
};
let mostPopularLang = '';
let maxCount = 0;

for (const lang in languageCounts) {
  if (languageCounts[lang] > maxCount) {
    maxCount = languageCounts[lang];
    mostPopularLang = lang;
  }
}
// console.log(`Most Popular:
${mostPopularLang}`);
```

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DYNAMIC UI DISPLAY

HTML Tables - Dynamically generated using DOM methods (document.createElement) and Template Literals.

Statistic Metric Cards - Automatically recalculates key performance indicators on every new submission:

- MOST POPULAR PROGRAMMING LANGUAGE
- TOP INTEREST AREA
- AVERAGE SATISFACTION RATING

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CONCLUSION

Successfully implemented a fully functional data-entry to data-visualization pipeline.

Gained deep practical understanding of DOM Manipulation.

Mastered core JavaScript data structures (Arrays and Objects) and iterative loops.

Built a mobile-responsive interface purely with standard CSS techniques.

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